ScienceDirect

Engineering Applications of Artificial Intelligence

Volume 144, 15 March 2025, 110145

Explainable artificial intelligence models in intrusion detection systems

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https://doi.org/10.1016/j.engappai.2025.110145

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Abstract

This study introduces the growing necessity of Explainable Artificial Intelligence (XAI) in the context of cybersecurity, particularly in Intrusion Detection Systems (IDSs). As Artificial Intelligence (AI) technologies, especially complex neural networks, become more prevalent and difficult to understand, the opacity of these models poses increasing risks in critical firald like cybersecurity. XAI aims to address these concerns by making AI models and their outputs more interpretable and transparent, thus fostering trust and secure use. Despite the rising interest and application of XAI in various domains, its integration into IDSs remains insufficient. This research provides a comprehensive overview of XAI, examining its definitions, terminologies, and the evolution of the field. It delves into the opportunities, challenges, and research needs within XAI, as well as the latest developments, tools, and technologies for implementing XAI in AI-based IDSs, alongside associated risks. The study also summarizes and demonstrates a gap in the literature regarding comprehensive reviews of IDS solutions that incorporate XAI, aiming to fill this gap by detailing the application of XAI in IDS research. The findings are expected to guide researchers, experts, users, and policymakers in the domains of AI, XAI, and IDS.

Introduction

There have been various security approaches and technologies to ensuring cyber and information security. While cryptographic approaches are generally used to protect information and provide confidentiality and integrity, many different security approaches and technologies such as firewalls, antiviruses, Denial of Service (DoS) mitigation solutions, and intrusion detection systems (IDS), have been used to protect information accessibility. In addition, developing new tools using the

most up-to-date techniques and configuring them according to customized rules and policies helps us prevent attacks to a large extent and ensure information security more. IDSs are software and hardware that enable us to take precautions against attacks on information systems, detect attacks, and report them by generating warnings or alarms. By using IDSs, unauthorized access and misuse of systems are detected and the damages they may cause are prevented as early and automatically as possible. IDSs have reached a market size of 5.6 billion dollars in 2022. The market size is estimated to reach 15.5 billion dollars, growing by over 11.5% every year until 2028 (Market Data Forecast). While IDS solutions had a share of over 64% in the market in 2018, IDSs are expected to dominate the market by 2025 (Global Market Insights). Having various advanced features such as data loss prevention, unauthorized access prevention, malicious activity detection, web application firewall and DoS prevention have enabled the market for IDSs to grow.

IDSs, especially equipped with developing technologies such as artificial intelligence (AI) and deep learning (DL), are becoming an important actor day by day in ensuring the security of information communication technology (ICT) systems. IDSs that use AI techniques, also referred to as intelligent IDSs, significantly contribute to ensuring cybersecurity as an increasingly popular and effective defense mechanism.

There has been a significant increase in the popularity and use of AI and ML over the past decade. AI continues to be used more widely and develops in many different areas of daily life. A study by Grand View Research evaluated the size of the global AI market at US$136.55 billion in 2022 and predicted a compound annual growth rate (CAGR) of 37.3% from 2023 to 2030 (Grand View Research). In the research conducted by Next Move Strategy Consulting, the size of the global explainable artificial intelligence (XAI) market was evaluated as 4.4 billion US dollars in 2021 and predicted to reach 21 billion US dollars with a compound a CAGR of 18.4% from 2022 to 2030 (Next Move Strategy Consulting). In parallel with these developments, the use of AI in IDSs is increasing significantly, and as a result, IDSs are developing more and more every day. IDSs are widely used and are becoming an indispensable part of cybersecurity strategies. As a result of this trend, a wide variety of new approaches and techniques have emerged that achieve surprising results, such as DL. These models, which have been introduced in recent years, have significantly improved compared to previous models in terms of prediction performance (Accuracy, Precision, Recall, F1-Score, etc.). However, there are several features of these models that have not received much attention, such as complexity, interpretability, and fairness. The increasing adoption of AI technologies, especially popular, incomprehensible, complex, and opaque ML models such as neural network models, is becoming increasingly difficult to understand. This makes the problem worse and more important in a field such as cybersecurity because trusting a system that cannot explain the reason for important decisions and leaving it alone poses obvious dangers. As a result, the complexity of current techniques is increasing, resulting in AI/ML black box models that cannot be easily examined or verified and are too complex for any human to comprehend. Such models can solve various complex problems, but they remain unreliable due to situations where predictions have negative effects. To overcome this problem, XAI proposes a transition to more interpretable, explainable, and understandable AI. XAI, a holistic approach to solutions, uses various methods to understand, comprehend, and interpret AI models. The field of research in XAI aims to solve the explainability and interpretability problem of AI and explain AI model's decision to users in terms that users can understand. It reveals more details about the behavior of complex ML models that

help experts validate model predictions. However, as AI is frequently used in high-risk areas such as cybersecurity, health, finance, transportation, autonomous driving, military, and law, the need for stakeholders to understand AI is increasing due to the dangerous consequences it may cause. This need for understanding also confronts researchers as a problem that must be overcome in order for people to trust and adopt AI technologies.

XAI approaches have been adopted to address the need for explainability in high-risk areas such as cybersecurity (Chen et al., 2021), healthcare (Tjoa and Guan, 2021a), finance (Silva et al., 2022), transportation (Zhang et al., 2020), autonomous driving (Lu et al., 2021), the military (Nguyen et al., 2020), and law (Dorr et al., 2021). Recently, XAI techniques have started to be used as a cybersecurity solution for IDSs, as well as in many other critical research areas. Cybersecurity researchers, users, policymakers, and all stakeholders in this field need to understand the importance of AI in order to fully benefit from its capabilities. XAI plays an important role in understanding AI models in the field of cybersecurity. The focus of this study differs from other studies as it highlights the latest developments and approaches to developing XAI applications and presents available tools and technologies for related topics and concepts, and provides a summary to elaborate the power of XAI models in IDSs. However, although some other studies exist on XAI and cybersecurity independently, no comprehensive study focuses on a broad review of IDS solutions based on XAI. Therefore, to fill this gap, this article provides a comprehensive review on XAI in IDSs for the last five years, highlights existing state-of-the-art work, the need for XAI, and the importance of XAI for IDSs.

Few studies on XAI have been published in the literature. Particularly in the field of cybersecurity, very few studies have been conducted. However, XAI and cybersecurity have been addressed separately in many studies. Additionally, there is no comprehensive research in the literature specifically focusing on the use of XAI in IDSs. Generally, it has been evaluated as a brief subheading in studies addressing cybersecurity. Fig. 1 shows the number of all XAI studies, XAI studies in the field of cybersecurity, and XAI studies in the field of IDSs in Google Scholar between 2018 and 2024. It can be seen from the figure that there are very few studies that cover both XAI and IDS domains together. For example, in 2023, the percentage of studies that include IDS among all XAI studies is approximately 3%. This study fills this gap in the literature by providing a comprehensive introduction to various XAI techniques applied in IDSs. Our study also offers a thorough background on XAI.

There have been many studies in the literature that aim to guide researchers by examining the use of AI in the field of cybersecurity. For example, Veiga (Veiga, 2018), by detailing the innovative solutions and advantages offered by AI in network security, demonstrate that AI can increase the effectiveness of existing security systems. The study provides important insights for better understanding the potential of AI in network security and for guiding future innovations in this field. In (Wickramasinghe et al., 2018), Wickramasinghe et al. examined the general applications of deep learning (DL) in ensuring the security of cyber-physical systems (CPS). They focused on how deep learning can be integrated into CPS security, existing approaches, and the generalizability of these approaches. As a detailed study on the use of AI in cybersecurity, Shaukat et al. (2020) provide a comprehensive review of ML techniques used in the field of cybersecurity over the past decade. This study explores how ML techniques are applied in cybersecurity and the advancements in this

field, showing that supervised and unsupervised learning techniques are used in many areas such as intrusion detection, malware detection, and fraud detection.

Due to the significant advances of AI-based IDSs in the field of cybersecurity, there are many studies that focus solely on examining and analyzing the use of AI in IDSs. For example, Mishra et al. (2019) focused on how ML techniques are used in IDSs, which are critical components of network security. The study analyzes in detail how ML techniques, particularly supervised, unsupervised, and semi-supervised learning, can be integrated into IDS systems, as well as the advantages and limitations of these approaches. It is demonstrated that ML methods can be successfully applied in IDSs, and these systems can automate and improve intrusion detection. Gumusbas et al. (Gumusbas et al., 2021) provide a comprehensive review of DL methods and databases for cybersecurity and IDS. Although IDSs are not covered in detail in the study, the datasets used in the application of DL methods are discussed more extensively. Another study on the use of AI in IDSs was conducted by Muneer et al. (Muneer et al., 2024). This study analyzes in detail how AI-based approaches are used in IDSs, emphasizing that AI is a powerful tool for enhancing the effectiveness of IDSs.

Due to the critical nature of decisions in the field of cybersecurity, many studies have emphasized the importance of investigating XAI from a cybersecurity perspective. Kuppa and Le-Khac (2021) examined how XAI methods can be used to increase the reliability of AI systems and how defenses can be developed against adversarial attacks. The study reveals the potential of XAI in cybersecurity while also highlighting the current challenges and future improvements in this area. Another study presented in (Srivastava et al., 2022) comprehensively covers the current status of XAI in cybersecurity, the challenges encountered, unresolved issues, and future research directions. Capuano et al. (2022) emphasize the importance of XAI in cybersecurity and analyze the existing techniques, their application areas, advantages, and limitations. The study also provides suggestions for future research to further promote the use of XAI in cybersecurity. Similarly, Fabien et al. (Fabien et al., 2022) conducted a comprehensive literature review on XAI in cybersecurity, examining how XAI is used in AI-based cybersecurity systems, its advantages, challenges, and future research directions. Another study on the use of XAI in cybersecurity was conducted by Zhang et al. (2022). This study provides an overview of the current state of research in this field, discussing the XAI techniques used in cybersecurity, the advantages these techniques offer, and the challenges they face. It also provides suggestions on how XAI can be further developed in cybersecurity. Like the aforementioned studies, Rjoub et al. (2023) provide a comprehensive discussion on the current state of XAI in cybersecurity and the latest developments in this field.

Due to the increasing need for explainability in many different critical application domains, numerous studies have been conducted that conceptually address explainability in a comprehensive manner and examine it in depth. In his study, Rudin (2019a) demonstrates why it is critical to prefer explainable models over black-box models and argues that these models should especially be preferred in high-risk decision processes by proposing approaches based on explainable models. He states that explainable models provide transparency, trust, and accuracy, while also facilitating auditing and error correction processes. Sahakyan et al. (2021) presented a comprehensive review on the use of XAI methods, particularly for tabular data. The study examined existing XAI techniques for tabular data, analyzing which methods were used, along with their advantages and limitations. Alicioglu and Sun (2021) presented a comprehensive review on how

XAI methods are supported by visual analytics. The study explores visual analytics techniques integrated with XAI methods, analyzing the advantages of these techniques, the challenges encountered, and potential future research areas. Ahmed et al. (2022) conducted a literature review examining how to make AI applications in Industry 4.0 explainable. It details which XAI techniques can be used to transform AI applications in Industry 4.0 to XAI, how they can be implemented, and the opportunities in this area. Kok et al. (2022) provided a comprehensive literature review on how XAI methods can be applied to make AI models used in the Internet of Things (IoT) more transparent. The study highlights the challenges XAI faces in IoT and suggests future research directions in this area. Chamola et al. (2023) presented a comprehensive review on Trustworthy and Explainable Artificial Intelligence. The study outlines why XAI is critical in the process of creating trustworthy AI, the components of trustworthiness, and how XAI can be used in AI systems. It particularly emphasizes the role of XAI in providing transparency, security, and fairness in fields such as healthcare, finance, law, and autonomous systems. Letrache and Ramdani (2023) conducted a comprehensive literature review and case study on model-agnostic XAI methods. The study introduces model-agnostic XAI methods, explains how these methods work, and examines their applicability to different AI models. It also demonstrates how these techniques can be applied in real-world scenarios through a case study.

Table 1 summarizes how our work is more comprehensive than other survey and review articles in the literature and highlights our contributions to these studies. The analysis of previous studies reveals our motivation and contributions to this research. While existing studies have analyzed AI applications in cybersecurity in detail, some have only examined XAI methods for a narrow set of cybersecurity applications. Additionally, some studies have not provided a detailed explanation of the background of XAI and cybersecurity. Upon examining Table 1 and it can be seen that this study is unique and distinct in terms of its detailed analysis of XAI and its comprehensive exploration of the relationship between IDS and XAI, compared to previously published studies in the field.
Black box models are ML models that cannot be understood by just looking at their parameters. Unlike linear models and decision trees, ML and especially DL models are often described as black box models due to their complexity. Black box models are inherently opaque. It can be difficult for even skilled experts to fully grasp these 'black box' models. Many ML and DL models essentially appear as “black boxes,” as shown in Fig. 2, which do not specify the details of decision in their internal mechanisms and predictions.

This lack of transparency and understanding in ML and DL algorithms can have serious consequences for trust and adoption in the models. For example, how can we know whether model predictions are wrong? This is especially vital in high-risk areas such as healthcare and cybersecurity. If a trained model has 99% accuracy, can a cybersecurity professional trust its attack prediction? What if the model misses the most important attack events? What if the high accuracy is due to data leakage in the test data, and out-of-sample performance is much worse? Because of questions like these, XAI is vital to the adoption of ML. It becomes necessary for ML models to be explainable for high-risk decisions such as finance, medical diagnosis, and cyber security (Kleinberg et al., 2018; Lakkaraju et al., 2019). The usefulness of black box models is becoming increasingly questionable, and some researchers argue that they should not be used for high-risk decisions (Rudin, 2019b).

This study fundamentally provides a holistic perspective on different aspects of XAI. It is a guide that shows those who are new to XAI or experts in the field what XAI is, what it aims for, what it can do, what its current situation is, where current trends, terms, and techniques can be interpreted effortlessly and aims to provide significant gains to those who are interested. However, after presenting general information about IDSs, a comprehensive study is presented that focuses on reviewing, classifying, and summarizing a wide range of IDS solutions based on XAI. In summary, this study contributes to researchers, experts, users, and policy makers in the field of cybersecurity, AI, and XAI as follows.

1) This study first provides an up-to-date and comprehensive analysis that emphasizes the importance of XAI for achieving trustworthy AI. It also introduces several key concepts, including a detailed definition of XAI, its terminology, objectives, current XAI toolboxes, and a thorough classification of XAI techniques, offering researchers a holistic perspective. In particular, it clarifies the interrelated concepts used in the literature as alternatives to explainability, which often lead to confusion.

2) Important criteria for evaluating the quality and meaningfulness of explanations produced by XAI techniques are presented, as well as indicating current research challenges, open problems and possible solutions.

3) The study rationalizes the motivations for integrating XAI into AI-based IDS models and reveals the opportunities, challenges, and further research needs for the use of XAI in IDSs.

4) Although there are some other studies in the literature on XAI and cybersecurity, there is no comprehensive study that extensively reviews IDS solutions based on XAI.

5) The study also addresses the relationship between the fields of XAI and LLM, which have gained increasing popularity in recent years. Evaluations were made regarding the mutual impact of these two important and evolving areas of study.

The aim of this study is to provide a comprehensive and up-to-date background on XAI and to examine the current status and studies in the literature on XAI applications in intrusion detection systems. For this purpose, the reference set was created by carefully selecting reputable articles and studies published in peer-reviewed conferences, journals, preprints, and websites, following the research methodology illustrated in Fig. 3. The following criteria were established for collecting the reviewed research articles.

1) First, the basic keywords “XAI” and “IDS” were used in search engines such as Google Scholar (Google scholar), IEEE Xplorer (IEEE Xplore), Springer (SpringerLink), and ScienceDirect (ScienceDirect). Additionally, search results were obtained using synonymous keywords like “Explainable Artificial Intelligence”, “Explainable AI”, “Explainable ML”, “Interpretable ML”, “Interpretable Artificial Intelligence”, “Interpretable AI”, and “Intrusion Detection System”.

2) Using the specified keywords, advanced search methods such as “Explainable AI” AND “Intrusion Detection System” OR “XAI” AND “IDS” were employed in the search engines, and results were obtained. The number of studies retrieved from each search engine is shown in Fig. 4. The selection criteria were defined as addressing both XAI and IDS together and being an IDS design study with explainability features.

3) Studies published in between 2018 and 2024 were selected to create a report on the latest trends in the application of XAI techniques in intrusion detection systems.

4) Academic search engines, many websites, such as Medium and GitHub, were reviewed, especially for current XAI techniques and toolboxes.

5) Publications written in English were only included in this study.

6) The citations of the reviewed studies were also tracked to ensure comprehensive coverage of relevant research.

7) The results were carefully filtered by the authors to highlight the most reputable studies. The titles and abstracts of the filtered studies were scanned and selected according to the selection criteria, and the chosen studies were thoroughly reviewed.

The study presents background information about XAI, and then studies on IDSs and the use of XAI in IDSs are included. Section 2 provides comprehensive information about the definition, classification, explanation types, and goals of explainable artificial intelligence. Section 3 presents information about the importance, difficulties and studies in the literature of using XAI in IDSs. Sections 4 Challenges, open issues and future directions, 5 Results and discussions discuss the issues, challenges, open issues, and discussions for using XAI in IDSs. Finally, the conclusions and future directions obtained in Section 6 are mentioned.

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Section snippets

Background knowledge of XAI approaches

XAI methods can provide explanations according to the stages of implementation of a model before, during or after the prediction as shown in Fig. 5. Ante-hoc methods involve external explanations before the training phase. Intrinsic methods refer to self-explanatory models that benefit from the internal structure of the model to provide explainability. Post-hoc methods refer to techniques that can be applied to any trained black box model without the need to understand the internal structure of …

XAI in intrusion detection systems

Intrusion refers to any unauthorized activity that disrupts or damages an information system. That is, any attack that violates the integrity, confidentiality or availability of an information system is defined as an intrusion. The purpose of IDS is to detect any abnormal activity in an information system that cannot be detected by basic security devices such as firewalls. Therefore, an IDS has become a necessary tool in computer networks to provide a more secure network environment.

IDSs are …

Challenges, open issues and future directions

This section discusses the insights and lessons learned from examining XAI methodologies for IDSs. Moreover, based on detailed reviews in this article, we discussed open issues and possibly highlighted research challenges for future directions. In this study, XAI techniques are discussed in detail. Almost all explainability approaches in the literature are presented in this study to guide the readers and find the answers to the research questions.

While XAI presents many conveniences and …

Results and discussions

As summarized in this article, XAI models offers solutions to IDS problems given above by creating more interpretable and understandable AI models. XAI employs various methods to help comprehend AI decisions, showing which data or data regions influenced these decisions. Despite the attention and application of XAI in many fields, its use in IDSs remains limited. IDSs, crucial for detecting data breaches and attacks, need high performance and the ability to justify their decisions. Applying XAI …

Conclusion and future directions

This study reviews XAI models and techniques from different perspectives, examining their applications and performances in IDSs. It provides detailed definitions and terminology, explores opportunities and challenges, and summarizes approaches, tools, datasets, and technologies for developing XAI models and applications, and also highlights the necessary steps for implementing XAI in AI-based IDSs and addresses the associated risks.

The results achieved in this study given in Table 1, Table 2, …

CRediT authorship contribution statement

Samed AL: Visualization, Methodology, Investigation, Conceptualization. **Seref Sagiroglu:** Writing – review & editing, Supervision, Methodology, Conceptualization. …

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

Acknowledgments

This work has been supported by Gazi University Scientific Research Projects Coordination Unit under grant number #FGA-2022-8043. …

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